

1.1 Conformal Mapping

The main feature of conformal mappings is that they are angle-preserving (except at some critical points) and allow a geometric approach to complex analysis.

Consider a complex function $w = f(z)$ defined in a domain D of the z -plane; then to each point in D there corresponds a point in the w -plane. In this way we obtain a mapping of D onto the range of values of in the w -plane.

Definition 1.1. *A mapping $w = f(z)$ is called conformal if it preserves angles between oriented curves in magnitude as well as in sense.*

The angle α ($0 \leq \alpha \leq \pi$) between two intersecting curves C_1 and C_2 is defined to be the angle between their oriented tangents at the intersection point z_0 . And *conformality* means that the images C_1^* and C_2^* of C_1 and C_2 make the same angle as the curves themselves in both magnitude and direction. See Fig.1.1

Theorem 1.1.1. *The mapping by an analytic function $w = f(z)$ is conformal, except at critical points, that is, points at which the derivative f' is zero.*

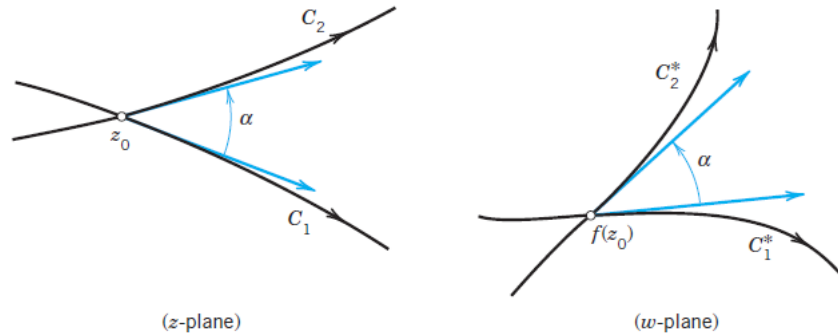


Fig. 1.1: Curves C_1 and C_2 and their respective images C_1^* and C_2^* and under a conformal mapping $w = f(z)$

1.1.1 The mapping $w = z^2$

Critical point of $w = z^2$ are given by $\frac{dw}{dz} = 0$.

$$\implies 2z = 0 \text{ or } z = 0.$$

Hence the transformation $w = z^2$ is conformal except at $z = 0$.

Now let us find the images of some basic curves under this mapping:

First, let us find the real and imaginary parts of $w = z^2$.

$$w = z^2 \implies u + iv = (x + iy)^2$$

$$\implies u = x^2 - y^2 \text{ and } v = 2xy.$$

In polar form, let $z = re^{i\theta}$ and $w = Re^{i\phi}$. Then, $Re^{i\phi} = r^2e^{i2\theta}$.

$$\implies R = r^2 \text{ and } \phi = 2\theta.$$

Images of some basic curves under this mapping:

1. The image of real axis ($y = 0$) in the z -plane.

$$\implies v = 0 \text{ and } u = x^2$$

$$\implies \text{positive real axis in the } w\text{-plane.}$$

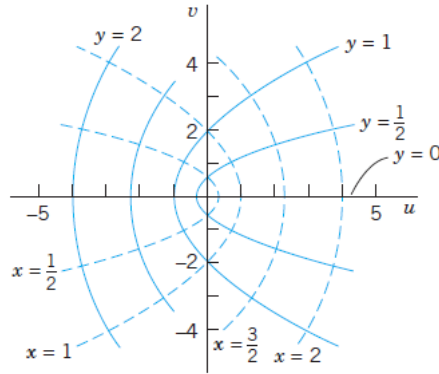


Fig. 1.2: Images of $x = \text{constant}$, $y = \text{constant}$ under $w = z^2$

2. The image of imaginary axis ($x = 0$) in the z -plane.

$$\implies v = 0 \text{ and } u = -y^2$$

$\implies$ negative real axis in the w -plane.

3. The image of lines parallel to the y -axis ($x = c$, c is a constant).

$$\implies u = c^2 - y^2 \text{ and } v = 2cy$$

Eliminating y , by putting $y = \frac{v}{2c}$ in u , we have $u = c^2 - \left(\frac{v}{2c}\right)^2$

$$\implies v^2 = 4c^4 - 4c^2u$$

$$\implies v^2 = -4c^2(u - c^2)$$

$\implies$ the images are parabolas with vertex at $(c^2, 0)$ and open to the left. See

Fig. 1.2.

4. The image of lines parallel to the x -axis ($y = c$, c is a constant).

$$\implies u = x^2 - c^2 \text{ and } v = 2cx$$

Eliminating x , by putting $x = \frac{v}{2c}$ in u , we have $u = \left(\frac{v}{2c}\right)^2 - c^2$

$$\implies v^2 = 4c^4 + 4c^2u$$

$$\implies v^2 = 4c^2(u + c^2)$$

$\implies$ the images are parabolas with vertex at $(-c^2, 0)$ and open to the right. See Fig.1.2.

1.1.2 The mapping $w = e^z$

Critical point of $w = e^z$ are given by $\frac{dw}{dz} = 0$.

$\implies e^z = 0$.

But $e^z \neq 0$ for all $z \in \mathbb{C}$. Hence the transformation $w = e^z$ is conformal at all z in the complex plane.

The real and imaginary parts of e^z are $u = e^x \cos y$ and $v = e^x \sin y$.

Images of some basic curves under this mapping:

1. The image of real axis ($y = 0$) in the z -plane.

$\implies v = 0$ and $u = e^x$

$\implies$ positive real axis in the w -plane.

2. The image of imaginary axis ($x = 0$) in the z -plane.

$\implies v = \sin y$ and $u = \cos y$

$\implies u^2 + v^2 = 1$

$\implies$ Hence the imaginary axis in the z -plane is transformed to a circle with centre $(0, 0)$ and radius 1 (unit circle) in the w -plane.

3. The image of lines parallel to the y -axis ($x = c$, c is a constant).

$\implies u = e^c \cos y$ and $v = e^c \sin y$

Eliminating y , $\cos y = \frac{u}{e^c}$ and $\sin y = \frac{v}{e^c}$

$$\implies \cos^2 y + \sin^2 y = \left(\frac{u}{e^c}\right)^2 + \left(\frac{v}{e^c}\right)^2$$

$$\implies u^2 + v^2 = e^{2c}$$

$\implies$ the images are circles with centre at $(0, 0)$ and radii e^c in the w -plane.

4. The image of lines parallel to the x -axis ($y = c$, c is a constant).

$$\implies u = e^x \cos c \text{ and } v = e^x \sin c$$

Eliminating x , we have $\frac{u}{\cos c} = \frac{v}{\sin c}$

$$\implies \frac{v}{u} = \tan c$$

$$\implies v = (\tan c)u$$

$\implies$ the images are straight lines through the origin in the w -plane.

1.1.3 The mapping $w = \frac{1}{z}$

The function $w = \frac{1}{z}$ is conformal for all z except $z = 0$, since $\frac{dw}{dz} = -\frac{1}{z^2}$ is defined for all z except $z = 0$.

To separate the real and imaginary parts:

$$w = \frac{1}{z} \implies z = \frac{1}{w} \implies x + iy = \frac{1}{u+iv}$$

$$\implies x + iy = \frac{(u-iv)}{(u+iv)(u-iv)} = \frac{u-iv}{u^2+v^2}$$

$$\implies x = \frac{u}{u^2+v^2}, \quad y = \frac{-v}{u^2+v^2}$$

Images of some basic curves under this mapping:

1. The image of real axis ($y = 0$) in the z -plane.

$$\implies v = 0$$

$\implies$ the image of real axis in the z -plane is the real axis in w -plane.

2. The image of imaginary axis ($x = 0$) in the z -plane.

$$\implies u = 0$$

$\implies$ the image of imaginary axis in the z -plane is the imaginary axis in w -plane.

3. The image of line parallel to real axis ($y = c$) in the z -plane.

$$\implies c = \frac{-v}{u^2+v^2}$$

$$\implies u^2 + v^2 + \frac{v}{c} = 0$$

$$\implies u^2 + (v + \frac{1}{2c})^2 = (\frac{1}{2c})^2$$

$\implies$ a line parallel to real axis ($y = c$) in the z -plane is transformed to circle with centre $(0, -\frac{1}{2c})$ and radius $\frac{1}{2c}$ in the w -plane.

4. The image of line parallel to imaginary axis ($x = c$) in the z -plane.

$$\implies c = \frac{u}{u^2+v^2}$$

$$\implies u^2 + v^2 - \frac{u}{c} = 0$$

$$\implies (u - \frac{1}{2c})^2 + v^2 = (\frac{1}{2c})^2$$

$\implies$ a line parallel to imaginary axis ($x = c$) in the z -plane is transformed to circle with centre $(\frac{1}{2c}, 0)$ and radius $\frac{1}{2c}$ in the w -plane.

5. The image of a straight line $ax + by + c = 0$ in the z -plane.

$$ax + by + c = 0 \implies a\frac{u}{u^2+v^2} + b\frac{-v}{u^2+v^2} + c = 0$$

$$\implies au - bv + c(u^2 + v^2) = 0$$

$$\implies c(u^2 + v^2) + au - bv = 0 \quad \text{--- (i)}$$

If $c \neq 0$, (i) represents a circle and if $c = 0$, (i) becomes $au - bv = 0$, a straight line passing through the origin with w -plane.

1.1.4 The mapping $w = \sin z$

The analytic function $w = f(z) = \sin z$ is conformal at all points z except where $f'(z) = 0$.

$$\text{Now, } f'(z) = 0 \implies \cos z = 0 \implies z = \pm(2n+1)\frac{\pi}{2}.$$

Thus, $w = \sin z$ is conformal at all points except at $z = \pm(2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$.

To separate the real and imaginary parts:

$$w = \sin z \implies u + iv = \sin(x + iy)$$

$$\implies u + iv = \sin x \cos iy + \cos x \sin iy = \sin x \cosh y + i \cos x \sinh y$$

Thus, $u = \sin x \cosh y$ and $v = \cos x \sinh y$

Images of some basic curves under this mapping:

1. The image of real axis ($y = 0$) in the z -plane.

$$\implies u = \sin x \text{ and } v = 0.$$

Since $|\sin x| \leq 1$, the image of the line $y = 0$ is the section of the real axis where $|u| \leq 1$ (i.e. $-1 \leq u \leq +1$) of the w -plane.

2. The image of imaginary axis ($x = 0$) in the z -plane.

$$\implies u = 0 \text{ and } v = \sinh y.$$

Since $\sinh y$ takes all real values, the image of the line $x = 0$ is the imaginary axis of the w -plane.

3. The image of line $x = c$ in the z -plane.

$$\implies u = \sin c \cosh y \text{ and } v = \cos c \sinh y$$

$$\implies \cosh y = \frac{u}{\sin c}, \sinh y = \frac{v}{\cos c}$$

We have $\cosh^2 y - \sinh^2 y = 1$.

$$\implies \left(\frac{u}{\sin c}\right)^2 - \left(\frac{v}{\cos c}\right)^2 = 1$$

$\implies$ the image of $x = c$ is a hyperbola with foci $(\pm 1, 0)$ and eccentricity, $e = \operatorname{cosec} c$.

4. The image of line $y = c$ in the z -plane.

$$\implies u = \sin x \cosh c \text{ and } v = \cos x \sinh c$$

$$\implies \sin x = \frac{u}{\cosh c} \text{ and } \cos x = \frac{v}{\sinh c}$$

$$\implies \left(\frac{u}{\cosh c}\right)^2 + \left(\frac{v}{\sinh c}\right)^2 = 1 \text{ is an ellipse.}$$

Question 1.1.1. Find the images of the curves $|z| = c$ and $\arg z = c$ under the mapping $w = z^2$.

Solution: Let $z = re^{i\theta}$ and $w = Re^{i\phi}$.

$$\therefore w = z^2 \implies Re^{i\phi} = r^2 e^{i2\theta}$$

$$\implies R = r^2, \phi = 2\theta$$

1. $|z| = c$ represents a circle in the complex plane centered at the origin with radius c .

$$\implies |z| = r = c$$

$$\implies R = r^2 = c^2$$

$\implies$ The resulting curve or image in the w -plane is also a circle centered at the origin with radius c^2 , given by $|w| = c^2$.

2. $\arg z = c$ represents a ray in the complex plane originating from the origin and making an angle of c with the positive real axis.

$$\implies \theta = c \text{ or } \phi = 2\theta = 2c$$

$\implies$ The resulting curve or image in the w -plane is also a ray originating from the origin, making an angle of $2c$ with the positive real axis.

Question 1.1.2. Find the image of the region $0 \leq x \leq 1, \frac{1}{2} \leq y \leq 1$ under the mapping $w = e^z$.

Solution: In polar form $w = Re^{i\phi}$. Given $w = e^z = e^x e^{iy}$.

$$\implies |w| = e^x \text{ and } \arg w = \phi = y$$

$\implies$ the image of z has magnitude e^x and argument y .

The vertical straight line $x = 0$ is mapped to the unit circle $|w| = e^0 = 1$. Also, the vertical straight line $x = 1$ is mapped to the circle with radius $|w| = e^1 = e$.

The horizontal straight lines $y = \frac{1}{2}$ is mapped to line with argument $\phi = \frac{1}{2}$ and $y = 1$ is mapped to line with $\phi = 1$.

Hence, the rectangle $0 \leq x \leq 1, \frac{1}{2} \leq y \leq 1$ is mapped on to the region bounded by the circles $|w| = 1, |w| = e$ and rays with arguments 1 and $\frac{1}{2}$.

Question 1.1.3. Find the image of the infinite strip $-2 \leq x \leq 2$ under the mapping $w = e^z$.

Solution: Given $w = e^z = e^x e^{iy}$. Therefore,

$x = -2 \implies w = e^{-2} e^{iy} \implies$ image of line $x = -2$ is a circle with radius e^{-2} .

$x = 2 \implies w = e^2 e^{iy} \implies$ image of line $x = 2$ is a circle with radius e^2 .

Hence the region $-2 \leq x \leq 2$ is mapped to the region between the circles with radii e^{-2} and e^2 including the boundaries.

Question 1.1.4. Find the image of the line $y - x + 1 = 0$ under the mapping $w = \frac{1}{z}$.

Solution: $w = \frac{1}{z} \implies z = \frac{1}{w}$

$\implies x + iy = \frac{1}{u + iv}$

$\implies x = \frac{u}{u^2 + v^2}$ and $y = \frac{-v}{u^2 + v^2}$

$y - x + 1 = 0$ is mapped to

$$\begin{aligned}\frac{-v}{u^2 + v^2} - \frac{u}{u^2 + v^2} + 1 &= 0 \\ -v - u + u^2 + v^2 &= 0 \\ u^2 - u + v^2 - v &= 0 \\ (u - \frac{1}{2})^2 + (v - \frac{1}{2})^2 &= \frac{1}{2}\end{aligned}$$

is a circle centered at $(\frac{1}{2}, \frac{1}{2})$ and radius $\frac{1}{\sqrt{2}}$.

Question 1.1.5. Show that under the transformation $w = z + \frac{1}{z}$, circles with a radius $r = 1$ are transformed onto the segment $-2 \leq u \leq 2$ of the u -axis, while circles with a radius $r \neq 1$ are transformed into ellipses. Also, find the points at which the mapping is not conformal.

Solution: In polar form, $z = re^{i\theta} = r(\cos \theta + i \sin \theta)$. Then, if $w = u + iv$, we have

$$u + iv = re^{i\theta} + \frac{1}{re^{i\theta}} = re^{i\theta} + \frac{1}{r}e^{-i\theta} = r(\cos \theta + i \sin \theta) + \frac{1}{r}(\cos \theta - i \sin \theta).$$

Separating the real and imaginary parts,

$$u = a \cos \theta, \quad v = b \sin \theta \tag{1.1}$$

where $a = r + \frac{1}{r}$, $b = r - \frac{1}{r}$.

If $r = 1$, $u = 2 \cos \theta$ and $v = 0$. Since $-1 \leq \cos \theta \leq 1$, $-2 \leq u \leq 2$. Thus, circle with radius 1 is mapped onto the segment $-2 \leq u \leq 2$ of the u -axis in w -plane.

If $r \neq 1$, then $\frac{u^2}{a^2} + \frac{v^2}{b^2} = 1$. Thus, circles $|z| = r \neq 1$ are mapped onto ellipses $\frac{u^2}{a^2} + \frac{v^2}{b^2} = 1$.

Now, the derivative of w is

$$w' = 1 - \frac{1}{z^2} = \frac{(z+1)(z-1)}{z^2}$$

The critical points of w are given by $w' = 0$.

$$\implies \frac{(z+1)(z-1)}{z^2} = 0$$

$$\implies z = \pm 1.$$

These are the points at which the mapping is not conformal.

1.1.5 Bilinear transformation

Definition 1.2. The transformation

$$w = \frac{az + b}{cz + d} \quad (ad - bc \neq 0)$$

where a , b , c , d are complex constants and $ad - bc \neq 0$, is called a bilinear transformation.

It is also known as *Mobius or linear fractional transformation*. This transformation is conformal only when $\frac{dw}{dz} \neq 0$, i.e., $\frac{ad-bc}{(cz+d)^2} \neq 0$, i.e., $ad - bc \neq 0$. The inverse $z = \frac{-dw+b}{cw-a}$ of the transformation is also a bilinear transformation.

Definition 1.3. The points which coincide with their transformation are called *fixed or invariant points of the transformation*.

In other words, the fixed points of the transformation $w = f(z)$ are obtained from the equation $z = f(z)$. As an example, the fixed points of $w = z^2$ and $w = z^3$ are the solutions of the equations $z = z^2$ and $z = z^3$ respectively.

Definition 1.4. If z_1, z_2, z_3, z_4 are distinct points taken in the order in which they are written, then the cross ratio of these points is defined as

$$(z_1, z_2, z_3, z_4) = \frac{(z_1 - z_2)(z_3 - z_4)}{(z_2 - z_3)(z_4 - z_1)}$$

The cross ratio will change if the order of the factors is changed.

Theorem 1.1.2. A cross ratio remains invariant under a bilinear transformation.

If $w = \frac{az+b}{cz+d}$, ($ad - bc \neq 0$) is a bilinear transformation and w_1, w_2, w_3, w_4 are the images of the points z_1, z_2, z_3, z_4 under this bilinear transformation, then

$$(w_1, w_2, w_3, w_4) = (z_1, z_2, z_3, z_4).$$

Question 1.1.6. Find a bilinear transformation which maps the points $z_1 = -2, z_2 = 0, z_3 = 2$ into the points $w_1 = 0, w_2 = i, w_3 = -i$.

Solution: Let w be the image of z under the required transformation. Since the cross ratio remains invariant under a bilinear transformation, we have

$$(w, w_1, w_2, w_3) = (z, z_1, z_2, z_3)$$

$$\implies (w, 0, i, -i) = (z, -2, 0, 2)$$

$$\implies \frac{(w-0)(i+1)}{(0-i)(-i-w)} = \frac{(z+2)(0-2)}{(-2-0)(2-z)}$$

$$\implies \frac{2w}{w+i} = \frac{z+2}{2-z}$$

$$\implies 2w(2-z) = (w+i)(z+2)$$

$$\implies w(2-3z) = i(z-2)$$

$$\implies w = i \frac{z+2}{2-3z}.$$

Question 1.1.7. Find a BLT which maps the points 2, i , -2 into the points 1, i , -1 .

Solution: $(w, 1, i, -1) = (z, 2, i, -2) \implies w = \frac{3z+2i}{iz+6}$.

Question 1.1.8. Find the BLT which sends the points $z = 0, 1, \infty$ into the points $w = -5, -1, 3$ respectively. What are the invariant points in this transformation?

Solution: We have $(w, -5, -1, 3) = (z, 0, 1, \infty)$

$$\implies \frac{(w+5)(-4)}{(-4)(3-w)} = \frac{z(1-\infty)}{-1(\infty-z)}$$

$$\implies \frac{w+5}{3-w} = z \frac{(1-\infty)}{(z-\infty)}$$

$$\text{Now } \frac{(1-\infty)}{(z-\infty)} = \lim_{t \rightarrow \infty} \frac{1-t}{z-t} = \lim_{t \rightarrow \infty} \frac{-1}{-1} = 1$$

$$\therefore \frac{w+5}{3-w} = z \text{ or } w = \frac{3z-5}{z+1}$$

To get the invariant point or fixed point, put $w = z$.

$$\implies z = \frac{3z-5}{z+1} \text{ or } z^2 - 2z + 5 = 0$$

$$\implies z = 1 \pm 2i.$$

Hence the invariant points are $1 + 2i$ and $1 - 2i$.

1.2 Problems:

1. Verify CR-conditions for the function: $f(z) = \frac{x-iy}{x^2+y^2}$.
2. Show that the function $f(z) = \sqrt{|xy|}$ is not regular (analytic) at the origin, even though CR- equations are satisfied at that point.
3. Show that $u = y^3 - 3x^2y$ satisfies Laplace's equation (or is a harmonic function).
Find its conjugate and the corresponding analytic function.
4. Find an analytic function whose imaginary part is $3x^2y - y^3$ and which vanishes

at $z = 0$.

5. Show that $u = e^x(x \cos y - y \sin y)$ satisfies Laplace's equation and determine the function v such that $u + iv$ is an analytic function and which takes the value e at $z = 1$.
6. Find the image of $x > 1, y > 0$ under the transformation $w = \frac{1}{z}$.
7. Find the image of the region $|z - \frac{1}{2}| < \frac{1}{2}$ under the transformation $w = \frac{1}{z}$.
8. Find the image of the region $-1 \leq x \leq 2, -\pi < y < \pi$ under the transformation $w = e^z$.
9. Discuss the transformation $w = \cos z$.
10. Show that under the transformation $w = \frac{1}{z}$ the circle $x^2 + y^2 - 6x = 0$ is transformed in to a straight line in the w -plane.
11. Given $u + v = \frac{x}{x^2 + y^2}$, find $f(z) = u + iv$ subject to the condition $f(1) = 1$ if $f(z)$ is analytic.